Experimental Economics:
Applications to Applied Microeconomics

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Office hours: Wednesdays, 230-4pm (Room IAB 1432).

Prerequisites: If Masters level, Quant 2 or equivalent. Open to all Economics and Sustainable Development PhD students.

OVERVIEW

Course Description: This course will provide the student with the necessary tools to be an avid consumer, and eventually a producer, of the experimental literature. Thus, it will provide a summary of recent experimental findings and detail how to gather and analyze data using experimental methods. It will align these methods to the interesting current debates in applied microeconomics.

The official text for the course: List, J.A. (2026) *Experimental Economics*.

Supplementary texts:

Glennerster, R., & Takavarasha, K. (2013). *Running Randomized Evaluations: A Practical Guide*. Princeton University Press.

Gerber, A., & Green, D. (2012). *Field Experiments: Design, Analysis, and Interpretation*.

Class Schedule: XX

Teaching Assistant:
XX For any questions please email the TA.

GRADING

Grading: You are required to fulfill the following assignments with corresponding grading weights.

Chapter Presentation (25%):

Each student will present the core concepts of the chapter from the course textbook. Chapters 5 through 19 are available for selection.

- Chapter Selection: Students will be randomly assigned a draft number, which determines the order in which they can choose their chapter.
- Presentation Format and Length: Presentations are individual and should last around 45 minutes.
- Provided Materials: A set of slides will be provided to assist with preparation. Students are encouraged to customize these slides to align with their presentation style.

Questions and Answers (20%):

Each week, the chapter presenter is required to create two questions, along with their corresponding solutions, based on the chapter's content. All students are required to answer the questions proposed by one of the presenters weekly. The presenter will evaluate the quality of the answers, while the TA will assess the quality of the questions. The two questions collectively contribute 10% to the final grade, and the answers to all questions throughout the quarter also contribute 10%.

Research Project Presentation (20%):

During final's week, the students are required to prepare a 15-minute presentation on an applied microeconomics project related to the methods covered in the course. You are not expected to present the finished product, but at minimum an idea and an outline of how that idea could be worked out.

Final Paper (35%):

Students are required to write-up either 1) a complete standalone short methods paper (~ 10 pages), 2) a short methods paper for the purpose of the class but that could potentially be a section of a larger paper or, 3) a comprehensive research proposal for a empirical paper that includes abstract, introduction, methodology/experimental design, and expected results. The write-up is due on XX, 2026. Paper submission is necessary to pass the course.

SCHEDULE

PART I Experimental Methods in Economics

Week 1:

Lecture (Date)

Introduction - Robert Metcalfe (RM)

PART II Designing Economic Experiments

Week 2:

Lecture (Date)

Chapter 2 A Primer on Economic Experiments - RM

Chapter 3 Internal Validity: Identification in Econ. Experiments - RM

Week 3:

Lecture (Date)

Chapter 4 Statistical Conclusion Validity: Estimation in Economic Experiments

Chapter 5 Optimal Experimental Design

Week 4:

Lecture (Date)

Chapter 6 Randomization Techniques

Chapter 7 Heterogeneity and Causal Moderation

PART III Violations of Exclusion Restrictions

Week 5:

Lecture (Date)

Chapter 8 Mediation: Exploring Relevant Mechanisms

Chapter 9 Experiments with Longitudinal Elements

Week 6:

Lecture (Date)

Chapter 10 Within-Subject Design

Chapter 11 SUTVA: Interference and Hidden Treatments

PART IV Building Scientific Knowledge

Week 7:
Lecture (Date)

Chapter 12 Observability: Non-Random Attrition

Chapter 13 Complete Compliance: One-Sided and Two-Sided Violations

PART V The Practical and Ethical Sides of Economic Experiments

Week 8:
Lecture (Date)

Chapter 14 Statistical Independence and Compromised Randomization

Chapter 15 Building Confidence in (and Knowledge from) Experimental Results

Week 9

Chapter 16 Generalizability and Scaling

Chapter 17 The Ethics of Economic Experiments

Week 10:
Lecture (Date)

Chapter 18 Pre-Treatment Administrative Responsibilities
Chapter 19 Incentives in Experiments

PART VI: Applicants to Environmental and Public Economics

Week 11:
Lecture (Date)

Designing Field Experiments in Environmental Economics - RM

Readings: Gosnell et al. (2020), Fairweather et al. (2024)

Week 12:
Lecture (Date)

Designing Field Experiments for Public Economics - RM

Readings: Bhargava et al. (2017), Hallsworth et al. (2017), Butera et al. (2022).

Week 13:
Lecture (Date)
Research Project Presentation

Week 14:
Lecture (Date)
Recap and Review

Extra readings:

Bhargava, S., Loewenstein, G., & Sydnor, J. (2017). Choose to lose: Health plan choices from a menu with dominated option. *Quarterly Journal of Economics*, 132(3), 1319-1372.

Butera, L., Metcalfe, R., Morrison, W., & Taubinsky, D. (2022). Measuring the welfare effects of shame and pride. *American Economic Review*, *112*(1), 122-168.

Fairweather, D., Kahn, M. E., Metcalfe, R. D., & Olascoaga, S. S. (2024). Expecting climate change: A nationwide field experiment in the housing market. NBER w33119.

Gosnell, G. K., List, J. A., & Metcalfe, R. D. (2020). The Impact of Management Practices on Employee Productivity: A Field Experiment with Airline Captains. *Journal of Political Economy*.

Hallsworth, M., List, J. A., Metcalfe, R. D., & Vlaev, I. (2017). The behavioralist as tax collector: Using natural field experiments to enhance tax compliance. *Journal of Public Economics*, 148, 14-31